



Image Processing Lab

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EXPERIMENT TITLE

Derivative-Based Edge Detection

Start Time:
4:16:27 PM

End Time:
4:17:42 PM

Total Time Spent:
00:01:14

Completed Operations:
1/5

Aim

To study derivative-based edge detection by applying first-order and second-order derivative operators to digital images and observe the resulting edge information.

Simulation Summary

A total of 1 derivative operator was applied to selected binary image patterns. first-order derivative operators (Sobel) were used to detect intensity transitions and object boundaries. The resulting edge maps were generated from the corresponding derivative responses, allowing the detected edge structure of each selected image to be observed.

Execution Order

1. Sobel

Sobel Operator

Derivative Order: First-order

Input Image

Plus (7×7)

0	0	0	1	0	0	0
0	0	0	1	0	0	0
0	0	0	1	0	0	0
1	1	1	1	1	1	1
0	0	0	1	0	0	0
0	0	0	1	0	0	0
0	0	0	1	0	0	0

Kernel(s)

Kernel Size: 3×3

Kernel X

-1	0	1
-2	0	2
-1	0	1

Kernel Y

-1	-2	-1
0	0	0
1	2	1

Final Edge Output

5×5

0	4	0	4	0
4	4	2	4	4
0	2	0	2	0
4	4	2	4	4
0	4	0	4	0

Observation

The output edge map clearly highlights the horizontal and vertical boundaries of the selected pattern, with stronger responses at regions of sharp intensity change.

